Week 2 Report: Self-Supervised Vision Transformers with Mini-DINO

1.DINO Architecture and Intuition

DINO (Self-Distillation with No Labels) is a self-supervised learning method where a **Student** network learns by matching the output of a **Teacher** network across different augmented crops of the same image. Instead of using a pre-trained teacher, both networks share the exact same architecture, and the teacher is dynamically updated using a moving average of the student's weights.

Our FullDINOModel consists of two parts:

1. **Backbone (TinyViT):** A lightweight Vision Transformer for 32×32 CIFAR-10 images (4×4 patch size, 192 embedding dimension, 6 layers, and 3 attention heads). It uses a [CLS] token to capture the image representation and positional encoding interpolation to handle both 32×32 global crops and 16×16 local crops.
2. **Projection Head:** A 3-layer MLP ($192 \rightarrow 256 \rightarrow 256 \rightarrow 128$) with GELU activations, followed by L_2 normalization and a weight-normalized linear layer that outputs a 1024-dimensional vector.

2.Student–Teacher Mechanism

Both the Student (g_{θ_s}) and Teacher (g_{θ_t}) share the exact same FullDINOModel structure and are initialized with identical weights. However, their roles during training are strictly asymmetric:

- Student Network: Receives all crops (both global and local views). Gradients are computed via backpropagation through the cross-entropy loss, and parameters θ_s are optimized using AdamW ($lr = 10^{-3}$, $weight_decay = 10^{-4}$).
- Teacher Network: Receives only global crops. Its parameters have requires_grad = False (stop-gradient operator) so backpropagation never flows through the teacher. Instead, the teacher acts as a stable, evolving

target generator that guides the student to learn high-level semantic invariance.

3.EMA, multi-crop, centering and sharpening

a).Exponential Moving Average (EMA) Teacher

To prevent the target distribution from shifting too rapidly and causing training collapse, the teacher weights θ_t are updated after each optimizer step as a momentum-based moving average of the student weights θ_s:

$$\theta_t \leftarrow \lambda\theta_t + (1 - \lambda)\theta_s$$

We use a momentum hyperparameter of $\lambda = 0.996$. This acts as temporal ensembling (similar to Polyak-Ruppert averaging), ensuring the teacher consistently produces higher-quality representations than the current student.

b).Multi-Crop Augmentation ("Local-to-Global" Correspondence)

For each input image in our 5,000-image CIFAR-10 subset, DataAugmentationDINO generates $V = 6$ views:

- 2 Global Crops (32×32): Sampled at scale $(0.5, 1.0)$ with random horizontal flips, color jitter, random grayscale, Gaussian blur, and solarization (on the second crop).
- 4 Local Crops (16×16): Sampled at scale $(0.15, 0.5)$ covering small regions of the image.

Because the teacher only sees global views while the student sees both global and local views, minimizing the cross-entropy loss between them forces the student to infer global context from local parts (local-to-global correspondence).

c).Teacher Centering and Sharpening

Without negative pairs, self-supervised models can easily collapse either by outputting the exact same dominant feature for every image (mode collapse) or by outputting a flat, equal probability across all 1024 dimensions (uniform collapse).

DINO prevents collapse using two simple operations on the Teacher's output:

- **Centering (prevents mode collapse):** A running average of the Teacher's outputs (updated with a 0.9 momentum rate) is subtracted from the

Teacher's scores before applying softmax. This stops any single dimension from taking over.
- **Sharpening (prevents uniform collapse):** Both networks divide their outputs by a temperature value before the softmax, using a much lower temperature for the Teacher (0.04) than the Student (0.1). This forces the Teacher's output probabilities to be sharp and confident rather than flat.

Finally, the DINO loss computes the cross-entropy between the Teacher's 2 global views and the Student's 5 other views, averaging the error across all 10 pairs per image.

4. Experimental Observations

4.1 Training Dynamics

We trained Mini-DINO on 5,000 CIFAR-10 images (batch size 64) for 10 epochs on a T4 GPU:

- **Initial Warm-up (Epochs 1–2):** The loss briefly increases from **6.7102** to **6.8147** as the centering vector adjusts from its initial zero state.
- **Steady Convergence (Epochs 2–10):** Once the center stabilizes and the EMA Teacher updates, the loss drops steadily every epoch, reaching **6.7212** at Epoch 5 and **6.5793** by Epoch 10.

4.2 Embedding Space Visualization (t-SNE)

We extracted the 192-dimensional [CLS] token embeddings from the Student backbone across 1,000 test images and projected them into 2D using t-SNE. Even without labels during training, images from the same CIFAR-10 classes group into distinct clusters, clearly separating similar categories (such as vehicles vs. animals).

4.3 Emergence of Semantic Attention Maps

Visualizing the [CLS] token attention weights from the Teacher's final Transformer block across its 3 heads shows clear object segmentation:

- **Foreground Focus:** On the test image (Class 9), **Head 1** and **Head 2** focus directly on the main body of the vehicle while ignoring the background.

- **Complementary Features: Head 3** highlights the outer edges and secondary parts of the object, showing that the ViT learns object boundaries automatically without labels.

5. Ablation Results: With EMA vs. Without EMA

To isolate the role of the Momentum Teacher, we ran a 5-epoch ablation study with use_ema=False (keeping the teacher weights frozen at their random initialization $\theta_t^{(0)}$) and compared it against the standard EMA training (use_ema=True, $\lambda = 0.996$).

Key Takeaways:

1. **Stagnation Without EMA:** When use_ema=False, the teacher remains a static random projection. After Epoch 2, the ablation loss flatlines entirely between **6.7331** and **6.7263** (a negligible change of 0.0068 over 4 epochs) because the target distribution never improves.
2. **Co-Evolution With EMA:** When use_ema=True, the teacher absorbs the student's learned features at each step ($\lambda = 0.996$), creating a virtuous cycle where a better student builds a better teacher, which in turn provides sharper, more semantically meaningful targets—driving the loss down steeply from **6.8147** (Epoch 2) to **6.7212** (Epoch 5) and **6.5793** (Epoch 10).